UGANDA REVISION HUB

S5 Starter Study Notes

Batch 1 • Online and offline revision

S5 • Mathematics

Functions and function notation

Learning objectives

- Explain the meaning of a function
- Evaluate a function for a given input
- Find a simple value of an unknown using function notation

Lesson notes

A function is a rule that assigns exactly one output to each allowed input. The notation f(x) means the value of function f when the input is x; it does not mean f multiplied by x.

To evaluate a function, substitute the input wherever x appears and simplify using the order of operations.

The domain is the set of allowed inputs. Some functions have restrictions, for example a denominator must not be zero.

Worked example

If $f(x) = 2x^2 - 3$, then $f(2) = 2(2^2) - 3 = 8 - 3 = 5$.

Revision questions

1 If $f(x)=3x+1$, find $f(4)$.

2 If $g(x)=x^2-5$, find $g(-2)$.

3 For $h(x)=1/(x-3)$, state the value of x that is not allowed.

Suggested answers

1 13

2 -1

3 $x = 3$

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S5 • Biology

Cell structure and specialised cells

Learning objectives

- Describe the roles of major cell structures
- Explain cell specialisation
- Relate structure to function in examples

Lesson notes

Cells contain structures that perform different functions. The nucleus contains DNA and helps regulate cell activities; mitochondria release usable energy through aerobic respiration; ribosomes build proteins; and the cell membrane controls exchange with the surroundings.

Cell specialisation means a cell has features adapted to a particular job. In multicellular organisms, specialised cells work together in tissues and organs.

A red blood cell has a biconcave shape and contains haemoglobin for transporting oxygen. A root hair cell has a long projection that increases surface area for absorbing water and mineral ions.

Worked example

The long root hair increases contact with soil particles, helping the plant absorb water and mineral ions efficiently.

Revision questions

1. What is the role of mitochondria?
2. Give one adaptation of a red blood cell for its function.
3. How does a root hair cell's shape help it?

Suggested answers

1. Release energy through aerobic respiration
2. Biconcave shape; haemoglobin; no nucleus in a mature mammalian red blood cell
3. Its long projection increases surface area for absorption.

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